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```
import pandas as pd
import numpy as np

from google.colab import files

# Upload dataset
uploaded = files.upload()

# Read uploaded CSV
filename = list(uploaded.keys())[0]
df = pd.read_csv(filename)

print("Dataset loaded successfully!")
print(df.head())
```

[Choose files](#) messy_stud...data (1).csv
messy_student_data (1).csv(text/csv) - 426 bytes, last modified: 19/09/2026 - 100% done
 Saving messy_student_data (1).csv to messy_student_data (1) (1).csv
 Dataset loaded successfully!

	Student_ID	Name	Age	Gender	City	Score	Department
0	1	Ananya	20	Female	Bengaluru	85	CSE
1	2	Rahul	NaN	M	Bangalore	78	ISE
2	3	Sneha	21	female	Mysore	NaN	ECE
3	4	Kiran	22	Male	Mysuru	91	CSE
4	5	Priya	20	F	Bengaluru	88	ISE

```
# =====
# TASK 1: DATA CLEANING & PREPROCESSING
# =====

# 1. Inspect dataset structure

print("Dataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nDataset Information:")
df.info()

print("\nFirst 5 Rows:")
display(df.head())

# 2. Identify missing values

print("\nMissing Values:")
print(df.isnull().sum())

# 3. Identify duplicate records

print("\nDuplicate Records:")
print(df.duplicated().sum())

# 4. Display statistical summary

print("\nStatistical Summary:")
display(df.describe(include="all"))
```

```
Dataset Shape:
(12, 7)

Column Names:
['Student_ID', 'Name', 'Age', 'Gender', 'City', 'Score', 'Department']

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Student_ID  12 non-null    int64
1   Name        12 non-null    object
2   Age         11 non-null    object
3   Gender      11 non-null    object
4   City        11 non-null    object
5   Score       11 non-null    object
6   Department  12 non-null    object
dtypes: int64(1), object(6)
memory usage: 804.0+ bytes
```

First 5 Rows:

	Student_ID	Name	Age	Gender	City	Score	Department
0	1	Ananya	20	Female	Bengaluru	85	CSE
1	2	Rahul	NaN	M	Bangalore	78	ISE
2	3	Sneha	21	female	Mysore	NaN	ECE
3	4	Kiran	22	Male	Mysuru	91	CSE
4	5	Priya	20	F	Bengaluru	88	ISE

```
Missing Values:
Student_ID    0
Name          0
Age           1
Gender        1
City          1
Score         1
Department    0
dtype: int64
```

```
Duplicate Records:
1
```

Statistical Summary:

	Student_ID	Name	Age	Gender	City	Score	Department
count	12.000000	12	11	11	11	11	12
unique	NaN	11	5	6	5	10	3
top	NaN	Rohan	20	Female	Bangalore	77	CSE
freq	NaN	2	3	3	3	2	4
mean	6.333333	NaN	NaN	NaN	NaN	NaN	NaN
std	3.366502	NaN	NaN	NaN	NaN	NaN	NaN
min	1.000000	NaN	NaN	NaN	NaN	NaN	NaN
25%	3.750000	NaN	NaN	NaN	NaN	NaN	NaN
50%	6.500000	NaN	NaN	NaN	NaN	NaN	NaN
75%	9.250000	NaN	NaN	NaN	NaN	NaN	NaN
max	11.000000	NaN	NaN	NaN	NaN	NaN	NaN

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```
# =====
# DATA CLEANING
# =====

# Make a copy of the dataset
cleaned_df = df.copy()

# 1. Remove duplicate records
```

```

cleaned_df = cleaned_df.drop_duplicates()

# 2. Handle missing values

for column in cleaned_df.columns:

    if cleaned_df[column].dtype == "object":

        cleaned_df[column] = cleaned_df[column].fillna(
            cleaned_df[column].mode()[0]
        )

    else:

        cleaned_df[column] = cleaned_df[column].fillna(
            cleaned_df[column].median()
        )

# 3. Remove extra spaces from text columns

for column in cleaned_df.select_dtypes(include="object").columns:

    cleaned_df[column] = cleaned_df[column].str.strip()

# 4. Display cleaned dataset

print("Cleaned Dataset:")
display(cleaned_df.head())

# 5. Check missing values after cleaning

print("\nMissing Values After Cleaning:")
print(cleaned_df.isnull().sum())

# 6. Check duplicate records after cleaning

print("\nDuplicate Records After Cleaning:")
print(cleaned_df.duplicated().sum())

```

Cleaned Dataset:

	Student_ID	Name	Age	Gender	City	Score	Department
0	1	Ananya	20	Female	Bengaluru	85	CSE
1	2	Rahul	20	M	Bangalore	78	ISE
2	3	Sneha	21	female	Mysore	69	ECE
3	4	Kiran	22	Male	Mysuru	91	CSE
4	5	Priya	20	F	Bengaluru	88	ISE

Missing Values After Cleaning:

```

Student_ID    0
Name          0
Age           0
Gender        0
City          0
Score         0
Department    0
dtype: int64

```

Duplicate Records After Cleaning:

```

0

```

```

from google.colab import files

files.download("cleaned_dataset.csv")

```

```

# =====
# COMPLETE DATA CLEANING
# =====

# Create a copy of the original dataset
cleaned_df = df.copy()

```

```

# 1. Remove duplicate records
cleaned_df = cleaned_df.drop_duplicates()

# 2. Remove extra spaces from text columns
for column in cleaned_df.select_dtypes(include="object").columns:
    cleaned_df[column] = cleaned_df[column].str.strip()

# 3. Standardize Gender values
cleaned_df["Gender"] = (
    cleaned_df["Gender"]
    .astype("string")
    .str.lower()
    .replace({
        "m": "Male",
        "male": "Male",
        "f": "Female",
        "female": "Female"
    })
)

# 4. Standardize City values
cleaned_df["City"] = (
    cleaned_df["City"]
    .astype("string")
    .str.lower()
    .replace({
        "bangalore": "Bengaluru",
        "mysore": "Mysuru"
    })
)

# 5. Convert Age and Score to numeric
cleaned_df["Age"] = pd.to_numeric(
    cleaned_df["Age"], errors="coerce"
)

cleaned_df["Score"] = pd.to_numeric(
    cleaned_df["Score"], errors="coerce"
)

# 6. Handle missing values
cleaned_df["Age"] = cleaned_df["Age"].fillna(
    cleaned_df["Age"].median()
)

cleaned_df["Score"] = cleaned_df["Score"].fillna(
    cleaned_df["Score"].median()
)

cleaned_df["Gender"] = cleaned_df["Gender"].fillna(
    cleaned_df["Gender"].mode()[0]
)

cleaned_df["City"] = cleaned_df["City"].fillna(
    cleaned_df["City"].mode()[0]
)

# 7. Display cleaned dataset
print("CLEANED DATASET")
display(cleaned_df)

# 8. Check remaining missing values
print("\nMissing Values After Cleaning:")
print(cleaned_df.isnull().sum())

# 9. Check duplicate records
print("\nDuplicate Records After Cleaning:")
print(cleaned_df.duplicated().sum())

```

CLEANED DATASET

	Student_ID	Name	Age	Gender	City	Score	Department
0	1	Ananya	20.0	Female	bengaluru	85.0	CSE
1	2	Rahul	21.0	Male	Bengaluru	78.0	ISE
2	3	Sneha	21.0	Female	Mysuru	82.0	ECE
3	4	Kiran	22.0	Male	mysuru	91.0	CSE
4	5	Priya	20.0	Female	bengaluru	88.0	ISE
5	6	Arjun	21.0	Male	Bengaluru	75.0	ECE
6	7	Divya	21.0	Female	bengaluru	82.0	CSE
7	8	Vikram	23.0	Male	Bengaluru	69.0	ISE
8	9	Megha	20.0	Female	Mysuru	95.0	CSE
9	10	Rohan	22.0	Male	mysuru	77.0	ECE
11	11	Pooja	21.0	Female	Bengaluru	82.0	ISE

Missing Values After Cleaning:

Student_ID 0

Name 0

Age 0

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Score 0

```
# Save the final cleaned dataset  
cleaned_df.to_csv("cleaned_dataset.csv", index=False)  
  
print("Cleaned dataset saved successfully!")
```

Cleaned dataset saved successfully!

```
from google.colab import files  
  
files.download("cleaned_dataset.csv")
```

```
from google.colab import files  
  
files.download("cleaned_dataset (4).csv")
```